SIMATS ENGINEERING
ASSESSMENT TOOL 2
Pseudocode Writing Task (Algorithm Design)

COURSE NAME: COMPUTER VISION WITH OPENCV

COURSE CODE: DSA02

COURSE OUTCOME COVERED

CO5: *Understand video processing - motion computation - 3D vision - geometry.* (BTL 6)

Assessment Tool Weightage: 20%

Pseudocode Writing Tasks: Design a clear and structured pseudocode for the given problem by organizing the solution into logical stages of a computer vision pipeline. Ensure proper use of control flow, modular steps, and justification of key operations to satisfy the given constraints.

1. Real-Time Face Detection and Recognition Pipeline A system must process a live video stream to:

A. Capture frames from a camera

B. Detect faces in each frame

C. Recognize faces using a pre-trained dataset

D. Display bounding boxes and labels in real time Constraints:

- Must operate continuously
- Should minimize processing delay
- Should handle multiple faces

Write detailed pseudocode for the complete pipeline using OpenCV concepts, including image acquisition, preprocessing, detection, recognition, and display. Ensure modular design and efficient looping structure.

ANS:

1. Real-Time Face Detection and Recognition Pipeline

Objective

Design a real-time computer vision pipeline that continuously captures video frames, detects multiple faces, recognizes them using a pre-trained dataset, and displays bounding boxes with identity labels.

Pseudocode

```
ALGORITHM RealTimeFaceRecognition

INPUT:
    Live camera stream
    Pre-trained face recognition dataset

OUTPUT:
    Live video with face bounding boxes and identity labels

BEGIN

    MODULE 1: INITIALIZATION

        Import OpenCV library

        Open camera using VideoCapture

        Load pre-trained face detection model
            face_detector ← Load Haar Cascade Classifier

        Load pre-trained face recognition model
            face_recognizer ← Load trained recognition model

        Load person labels/names from the dataset

        Set detection parameters
            scale_factor ← 1.1
            min_neighbors ← 5

        Set recognition threshold

        IF camera cannot be opened THEN
            Display "Camera Error"
            STOP
        END IF


    MODULE 2: CONTINUOUS FRAME ACQUISITION
```

```
WHILE camera is running DO

    Capture one frame from camera

    IF frame is not successfully captured THEN
        CONTINUE
    END IF


    MODULE 3: PREPROCESSING

        Resize frame to smaller resolution
            frame ← Resize(frame)

        Convert frame from BGR to grayscale
            gray ← ConvertToGrayscale(frame)

        Apply optional histogram equalization
            gray ← EqualizeHistogram(gray)


    MODULE 4: FACE DETECTION

        Detect all faces in the grayscale frame

        faces ← DetectFaces(
                    gray,
                    scale_factor,
                    min_neighbors
                )

        FOR each face in faces DO

            Get bounding box coordinates
                x, y, width, height ← face

            Extract face region
                face_ROI ← gray[y : y+height,
                                x : x+width]
```

```
        MODULE 5: FACE RECOGNITION

            Resize face_ROI if required

            predicted_label,
            confidence ←
               RecognizeFace(face_ROI)

            IF confidence satisfies recognition threshold THEN
               name ← GetName(predicted_label)
            ELSE
               name ← "Unknown"
            END IF


        MODULE 6: DISPLAY RESULT

            Draw bounding rectangle around face

            Display person's name above
            or near the bounding box

            Display confidence if required

      END FOR


   MODULE 7: DISPLAY FRAME

      Show processed frame on screen

      Wait for a very short time

      IF user presses 'q' THEN
         BREAK
      END IF

  END WHILE


MODULE 8: TERMINATION
```

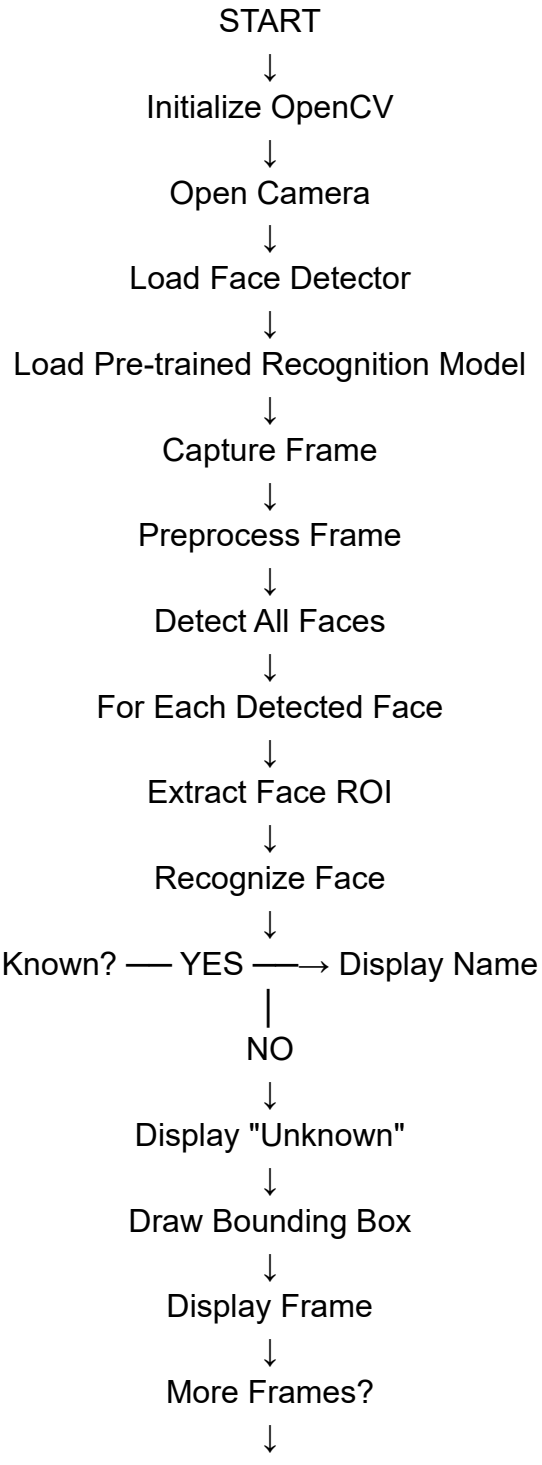
Release camera

Close all OpenCV windows

Display "Face Recognition Stopped"

END

Modular Pipeline



YES → Repeat

↓

NO / Press 'q'

↓

Release Camera

↓

END

Handling Multiple Faces

The algorithm uses a loop to process every detected face in the current frame.

FOR each face in detected_faces DO

Extract face ROI

Recognize face

Draw bounding box

Display corresponding name

END FOR

Therefore, if three faces are present, all three faces are detected, recognized independently, and displayed with separate labels.

Efficiency Considerations

1. **Continuous loop:** The WHILE loop continuously processes live camera frames.
2. **Frame resizing:** Smaller frames reduce processing time.
3. **Grayscale conversion:** Reduces computational requirements.
4. **ROI processing:** Recognition is performed only on detected face regions.
5. **Pre-trained models:** Avoids training during real-time operation.
6. **Short display delay:** Allows continuous frame processing.
7. **Multiple-face loop:** Processes all detected faces in the same frame.
8. **Immediate termination:** Pressing q safely stops the system and releases resources.

OpenCV Functions Used

Pipeline Stage	**OpenCV Concept / Function**
Image acquisition	cv2.VideoCapture()
Resizing	cv2.resize()
Grayscale conversion	cv2.cvtColor()
Contrast enhancement	cv2.equalizeHist()
Face detection	Haar Cascade detectMultiScale()
Face recognition	LBPH / pre-trained recognizer

Pipeline Stage	OpenCV Concept / Function
Bounding box	cv2.rectangle()
Label display	cv2.putText()
Video display	cv2.imshow()
Key detection	cv2.waitKey()
Resource release	cap.release()
Window closing	cv2.destroyAllWindows()

Conclusion

The proposed pseudocode divides the face recognition system into modular stages: initialization, image acquisition, preprocessing, face detection, face recognition, and display. The continuous WHILE loop ensures real-time operation, while resizing and grayscale conversion reduce processing delay. The FOR loop allows multiple faces to be processed in every frame. Finally, bounding boxes and identity labels are displayed continuously until the user presses q.

2. OCR Pipeline for Document Processing

A document processing system must:

A. Read input images containing printed text
B. Enhance image quality (noise removal, contrast improvement)
C. Segment text regions
D. Extract and display recognized text Constraints:

- Input images may be noisy and low contrast
- Batch processing required

Write pseudocode for an end-to-end OCR pipeline using OpenCV functions. Clearly structure preprocessing, segmentation, and text extraction stages.

ANS:

OCR Pipeline for Document Processing

Objective

Design an end-to-end OCR pipeline that reads printed text from input images, improves image quality, segments text regions, extracts recognized text, and processes multiple documents efficiently.

Pseudocode

ALGORITHM OCRDocumentProcessing

```
INPUT:
   Folder containing document images

OUTPUT:
   Recognized text from each document

BEGIN

   MODULE 1: INITIALIZATION

      Import OpenCV library
      Import OCR library

      Get list of all images from input folder

      IF input folder is empty THEN
         Display "No images found"
         STOP
      END IF


   MODULE 2: BATCH PROCESSING

      FOR each image_file in image_folder DO

         Display "Processing:", image_file

         Read input image
            image ← cv2.imread(image_file)

         IF image is not successfully loaded THEN
            Display "Image loading error"
            CONTINUE
         END IF


         MODULE 3: IMAGE PREPROCESSING

            Resize image to improve character visibility
               image ← cv2.resize(image)

            Convert image to grayscale
```

```
    gray ← cv2.cvtColor(
                image,
                BGR_TO_GRAY
            )

  Remove image noise
    gray ← cv2.GaussianBlur(
                gray,
                kernel_size,
                sigma
            )

  Improve contrast
    enhanced ← ApplyHistogramEqualization(gray)

  IF illumination is uneven THEN
    enhanced ← ApplyCLAHE(enhanced)
  END IF


MODULE 4: TEXT SEGMENTATION

  Convert enhanced image into binary image
    binary ← cv2.threshold(
                enhanced,
                OTSU
              )

  IF document has uneven lighting THEN
    binary ← cv2.adaptiveThreshold(
                enhanced
              )
  END IF

  Apply morphological operations
    binary ← MorphologicalOpening(binary)
    binary ← MorphologicalClosing(binary)

  Detect contours
    contours ← cv2.findContours(binary)

  Identify possible text regions
```

```
    FOR each contour in contours DO

        Calculate contour area

        IF contour area is greater than
          minimum_area THEN

            Create bounding box
              x, y, width, height ←
                cv2.boundingRect(contour)

            Store text region

        END IF

    END FOR


MODULE 5: TEXT REGION EXTRACTION

    FOR each detected text region DO

        Extract Region of Interest (ROI)

        Crop the text region from binary image

        Resize ROI if required

        Store ROI for OCR processing

    END FOR


MODULE 6: OCR TEXT EXTRACTION

    recognized_text ← EMPTY

    FOR each text ROI DO

        Send ROI to OCR engine
```

text ← OCR(ROI)

Add text to recognized_text

END FOR

MODULE 7: POST-PROCESSING

Remove unnecessary spaces

Remove unwanted characters

Correct obvious OCR formatting errors

Arrange recognized text in proper reading order

MODULE 8: OUTPUT

Display recognized_text

Save recognized_text to output file

Display "Processing completed"

END FOR

MODULE 9: TERMINATION

Display "All documents processed"

END

Complete OCR Pipeline

START

↓

Load Input Image Folder

↓

Read Image

↓

Resize Image

↓

Convert to Grayscale

↓

Noise Removal

↓

Contrast Enhancement

↓

Illumination Correction

↓

Thresholding

↓

Morphological Processing

↓

Text Region Segmentation

↓

Extract Text ROIs

↓

OCR Recognition

↓

Post-Processing

↓

Display Recognized Text

↓

Save Text

↓

Next Image

↓

All Images Processed?

↓

YES

↓

END

OpenCV Functions Used

Stage	**OpenCV Function / Concept**
Read image	cv2.imread()
Resize	cv2.resize()
Grayscale conversion	cv2.cvtColor()
Noise removal	cv2.GaussianBlur() / cv2.medianBlur()
Contrast enhancement	cv2.equalizeHist()

Stage	OpenCV Function / Concept
Local enhancement	cv2.createCLAHE()
Thresholding	cv2.threshold()
Adaptive thresholding	cv2.adaptiveThreshold()
Morphological processing	cv2.morphologyEx()
Text segmentation	cv2.findContours()
Region extraction	Array slicing / ROI
OCR	Tesseract or another OCR engine

Handling the Constraints

1. Noisy Images

Gaussian or median filtering removes small noise and unwanted pixels before OCR.

2. Low Contrast

Histogram equalization or CLAHE improves the contrast between text and background.

3. Uneven Illumination

Adaptive thresholding applies different thresholds to different image regions, making it suitable for documents with shadows or uneven brightness.

4. Batch Processing

The FOR loop processes every image in the input folder sequentially. This avoids loading all images into memory at the same time.

5. Efficient Processing

Only the detected text regions are passed to the OCR engine. This reduces unnecessary computation and improves processing speed.

Conclusion

The proposed OCR pipeline consists of image acquisition, resizing, grayscale conversion, noise removal, contrast enhancement, thresholding, morphological processing, text segmentation, OCR recognition, and post-processing. The system handles noisy and low-contrast documents using filtering and enhancement techniques. Adaptive thresholding improves performance under uneven illumination, while batch processing allows multiple documents to be processed efficiently with controlled memory usage.

3. Motion Detection and Tracking System A surveillance system must:

A. Read video input

B. Detect motion between consecutive frames

C. Track moving objects

D. Highlight detected motion regions Constraints:

- Should handle dynamic background
- Must operate in near real-time

Design pseudocode for the system, including frame differencing, filtering, tracking logic, and visualization. Ensure proper control flow and handling of continuous video streams.

ANS:

1. Motion Detection and Tracking System

Objective

Design a motion detection and tracking system that continuously reads a video stream, detects motion between consecutive frames, tracks moving objects, and highlights motion regions. The system should handle dynamic backgrounds and operate in near real-time.

Pseudocode

```
ALGORITHM MotionDetectionAndTracking

INPUT:
   Live video stream or video file

OUTPUT:
   Video with highlighted moving objects and tracking information

BEGIN

   MODULE 1: INITIALIZATION

      Import OpenCV library

      Open video source
         video ← cv2.VideoCapture(input_source)

      Create background subtractor
         background_model ←
            cv2.createBackgroundSubtractorMOG2(
               history = 500,
               varThreshold = 50,
               detectShadows = TRUE
            )

      Initialize object tracker
         tracker ← InitializeTracker()
```

```
Set minimum contour area
    MIN_AREA ← 500

IF video cannot be opened THEN
    Display "Video input error"
    STOP
END IF


MODULE 2: CONTINUOUS VIDEO PROCESSING

    WHILE video is running DO

        Read current frame
            success, frame ← video.read()

        IF success = FALSE THEN
            BREAK
        END IF


        MODULE 3: FRAME PREPROCESSING

            Resize frame if required
                frame ← cv2.resize(frame, desired_size)

            Convert frame to grayscale
                gray ← cv2.cvtColor(
                            frame,
                            BGR_TO_GRAY
                        )

            Apply Gaussian filtering
                filtered ← cv2.GaussianBlur(
                                gray,
                                (5,5),
                                0
                            )


        MODULE 4: FRAME DIFFERENCING
```

```
    Compare current frame with previous frame

    IF previous_frame does not exist THEN
        previous_frame ← filtered
        CONTINUE
    END IF

    frame_difference ← cv2.absdiff(
                          previous_frame,
                          filtered
                       )

    Apply threshold
        motion_mask ← cv2.threshold(
                          frame_difference,
                          25,
                          255,
                          BINARY
                      )

    previous_frame ← filtered


MODULE 5: DYNAMIC BACKGROUND HANDLING

    Apply adaptive background subtraction

        foreground_mask ←
            background_model.apply(frame)

    Combine or select the appropriate
    motion mask and foreground mask

    Remove shadows if required


MODULE 6: NOISE FILTERING

    Apply morphological opening
        motion_mask ←
            MorphologicalOpening(motion_mask)
```

```
Apply morphological closing
    motion_mask ←
        MorphologicalClosing(motion_mask)

Apply dilation to connect
nearby motion regions

    motion_mask ←
        cv2.dilate(
            motion_mask,
            kernel,
            iterations = 2
        )


MODULE 7: MOTION REGION DETECTION

    Find contours in motion mask

        contours ← cv2.findContours(
                        motion_mask
                    )

    FOR each contour in contours DO

        Calculate contour area

        IF contour area < MIN_AREA THEN
            Ignore contour
        ELSE

            Calculate bounding box

                x, y, width, height ←
                    cv2.boundingRect(contour)

            Store motion region

        END IF

    END FOR
```

```
MODULE 8: OBJECT TRACKING

    IF new motion regions are detected THEN

        FOR each detected region DO

            Initialize or update tracker
            using bounding box

        END FOR

    ELSE

        Update existing trackers

    END IF

    FOR each tracked object DO

        Update object position

        Assign object ID

        Store current position

    END FOR


MODULE 9: VISUALIZATION

    FOR each tracked object DO

        Draw bounding rectangle

        Draw object ID

        Draw tracking path if required

    END FOR

    Display motion mask
```

Display processed video frame

Display "Motion Detected"
when moving objects are present

MODULE 10: REAL-TIME CONTROL

Show processed frame

Wait for a very short time

IF user presses 'q' THEN
BREAK
END IF

END WHILE

MODULE 11: TERMINATION

Release video source

Destroy all OpenCV windows

Display "Motion Detection Stopped"

END

Frame Differencing Logic

Previous Frame
↓
Current Frame
↓
Absolute Difference
↓
Thresholding
↓
Motion Mask
↓
Noise Filtering
↓

Motion Regions

The absolute difference between consecutive frames identifies pixels that have changed. A threshold converts these changes into a binary motion mask.

Object Tracking Logic

Motion Region Detected

↓

Create Bounding Box

↓

Assign Object ID

↓

Track Object in Next Frames

↓

Update Position

↓

Object Still Visible?

↓ ↓

YES NO

↓ ↓

Continue Tracking Remove Tracker

Tracking prevents the system from treating the same moving object as a completely new object in every frame.

Dynamic Background Handling

Dynamic backgrounds such as moving trees, rain, shadows, and changing illumination can produce false motion. To handle this, an adaptive background subtraction method such as MOG2 is used.

Video Frame

↓

Adaptive Background Model

↓

Foreground Mask

↓

Noise Filtering

↓

Moving Object Detection

The background model continuously adapts to gradual background changes, reducing false detections.

Complete Pipeline

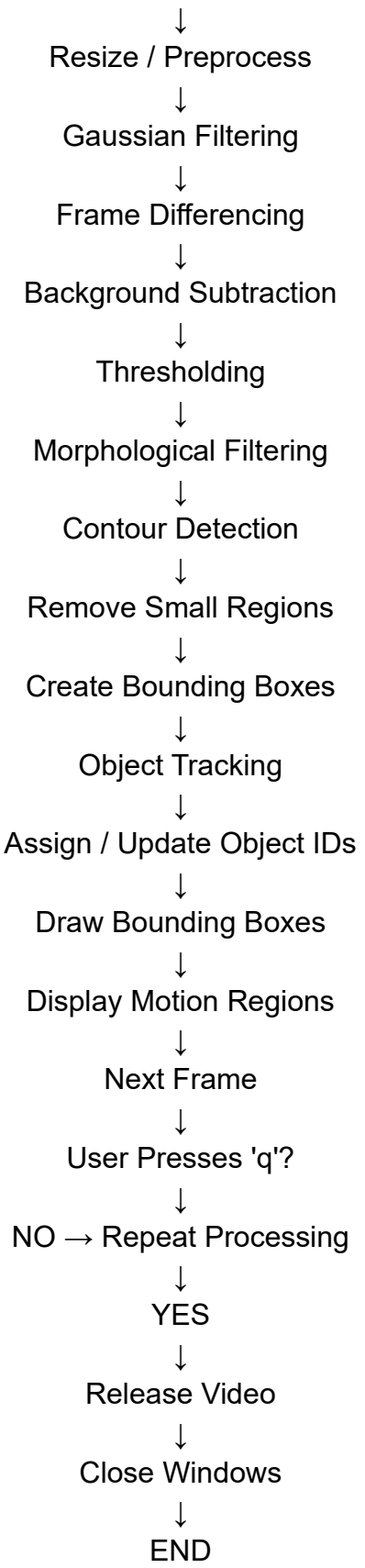
START

↓

Open Video Input

↓

Read Current Frame


Resize / Preprocess
Gaussian Filtering
Frame Differencing
Background Subtraction
Thresholding
Morphological Filtering
Contour Detection
Remove Small Regions
Create Bounding Boxes
Object Tracking
Assign / Update Object IDs
Draw Bounding Boxes
Display Motion Regions
Next Frame
User Presses 'q'?
NO → Repeat Processing
YES
Release Video
Close Windows
END

Handling the Constraints

Requirement	Solution
Read video input	cv2.VideoCapture()
Detect motion	Frame differencing using cv2.absdiff()
Dynamic background	MOG2 adaptive background subtraction
Remove noise	Gaussian blur and morphological operations
Remove false detections	Minimum contour-area filtering
Track moving objects	Object tracker with bounding boxes and IDs
Highlight motion	cv2.rectangle() and tracking paths
Near real-time	Lightweight OpenCV operations and frame resizing
Continuous video	WHILE loop processes frames continuously

Conclusion

The proposed motion detection and tracking system continuously reads video frames and detects changes using frame differencing. Gaussian filtering and morphological operations reduce noise, while MOG2 background subtraction handles dynamic backgrounds. Contours are used to identify significant motion regions, and object tracking maintains the position and identity of moving objects across frames. Bounding boxes, object IDs, and motion masks provide clear visualization. The lightweight processing pipeline allows the system to operate in near real time.

Code of Conduct:

I Amalraj P (192524089) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Problem Understanding	20%	Identifies all inputs, outputs, constraints accurately	Minor missing elements	Partial understanding	Incorrect interpretation
Algorithm Design	30%	Complete, correct, and logically sound solution	Minor logical gaps	Partially correct logic	Incorrect approach
Use of Control Structures	20%	Efficient and appropriate use of loops, conditions, functions/modules	Minor inefficiencies	Limited or basic usage	Incorrect or missing constructs
Pseudocode Structure	10%	Well-structured, clear, consistent format, easy to follow	Mostly clear with minor issues	Difficult to follow in parts	Poorly structured and unclear

Completeness	20%	Handles all cases; optimized approach	Handles most cases; reasonable efficiency	Handles basic cases only	Incomplete or inefficient